# SOLUTIONS - Exercise - Week 12: MATH1061/7861

(1) There are 5 females and 4 males on a list. Our goal is to choose one team of 3 people from this list.

(a) **(1 mark)** How many different teams can be chosen from the list?
Give your answer as an integer.

There are 9 possible people to choose from, so there are

$$\binom{9}{3} = \frac{9!}{6! \cdot 3!} = \frac{9 \cdot 8 \cdot 7}{6} = 84$$

different possibilities.

**MARKING:**
1 mark for providing the correct answer

(b) **(2 marks)** If there is to be at least one person of each gender (male or female) on the team, how many different teams can be chosen from the list?
Give your answer as an integer.

We count the number of teams that *do not* have at least one male and at least one female, i.e. the teams that are either all male or all female. This is

$$\binom{5}{3} + \binom{4}{3} = \frac{5 \cdot 4 \cdot 3}{6} + \frac{4 \cdot 3 \cdot 2}{6} = 10 + 4 = 14.$$

Subtracting this from the total found in part (a), there are $84 - 14 = 70$ possible teams with at least one male and at least one female.

**MARKING:**
1 mark for identifying a correct strategy (e.g. subtracting away the number of teams that are all male or all female)
1 mark for providing the correct answer

(c) **(2 marks)** What is the probability that a team chosen at random contains both a male and a female?
Express your answer as a rational number $\frac{a}{b}$ with $\gcd(a, b) = 1$.

Using the answers from (a) and (b), the probability is $\frac{70}{84} = \frac{5}{6}$.

**MARKING:**
2 marks for providing the correct answer

(d) **(2 marks)** What is the probably that a team chosen at random contains only males?
Express your answer as a rational number $\frac{a}{b}$ with $\gcd(a, b) = 1$.

There are $\binom{4}{3} = 4$ teams that contain only males, so the probability is $\frac{4}{84} = \frac{1}{21}$.

**MARKING:**
1 mark for computing the number of teams that only contain males
1 mark for providing the correct answer

(2) **(3 marks)** What is the coefficient of $x^3$ in the expansion of $(2x-3)^6$?

From the binomial theorem,

$$(2x-3)^6 = \sum_{k=0}^{6} \binom{6}{k} (2x)^k (-3)^{6-k} = \sum_{k=0}^{6} \binom{6}{k} 2^k (-3)^{6-k} x^k.$$

Looking at $x^3$, we need $k = 3$, so the coefficient is

$$\binom{6}{3} 2^3 (-3)^{6-3} = 20 \times 8 \times -27 = -4320.$$

**MARKING:**

1 mark for stating the binomial theorem (either explicitly, or implicitly by using it)
1 mark for substituting correctly into the binomial theorem
1 mark for providing the correct answer
Provide full marks if the students obtain the correct answer by expanding directly.